AAAI Fall Symposium: Plan4Planet

November 6, 2026

Planning Wildlife Monitoring Under Ecological Uncertainty

Scenario-Based Sensor Placement for Resource-Constrained Conservation Programs

Hannah Murray (USC), Amrita Gupta (Microsoft), Arielle W. Parsons (Conservation Science Partners), Justin P. Suraci (Conservation Science Partners), Bistra Dilkina (USC)

USC Center for AI in Society

Conservation Science Partners

Allen Family Philanthropies

Why do we need to monitor our wildlife?

Biomonitoring is the repeated measurement of species and ecosystems over time. It shows how ecosystems are changing.

Global commitments rely on monitoring data to report progress on the United Nations' Sustainable Development Goals.

Camera traps are one of the most widely used monitoring tools due to their passive, non-invasive nature.



Next-Generation Camera Trapping: Systematic Review of Historic Trends Suggests Keys to Expanded Research Applications in Ecology and Conservation

Zackary J. Delisle, Elizabeth A. Flaherty, Mackenzie R. Nobbe, Cole M. Wzientek and Robert K. Swihart*

Department of Forestry and Natural Resources, Purdue University, West Lafayette, IN, United States

Camera trapping is an effective non-invasive method for collecting data on wildlife species to address questions of ecological and conservation interest. We reviewed 2,167 camera trap (CT) articles from 1994 to 2020. Through the lens of technological diffusion, we assessed trends in: (1) CT adoption measured by published research output, (2) topic, taxonomic, and geographic diversification and composition of CT applications, and (3) sampling effort, spatial extent, and temporal duration of CT studies. Annual publications of CT articles have grown 81-fold since 1994, increasing at a rate of 1.26

What can camera traps tell us?

Goal

Ecologists use **spatial capture-recapture (SCR) models** to estimate species population size using images from camera traps.

Challenge

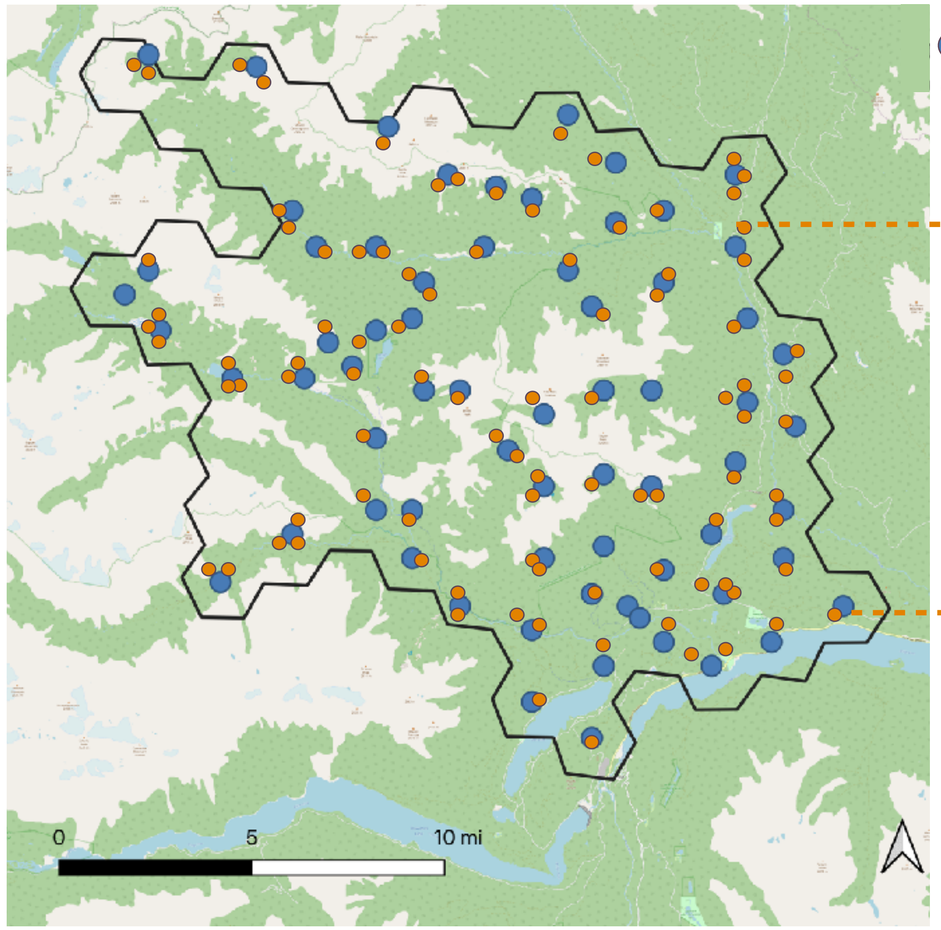
SCR models are influenced by the camera layout. Current methods do not consider species preferences, which **can lead to biased density estimates.**

Our Objective

Develop algorithmic strategies that account for species preferences to deploy sensors that minimize the error between predicted and true population size under budget.



Camera Trapping in Biodiversity Surveys



Trap
Species Observation

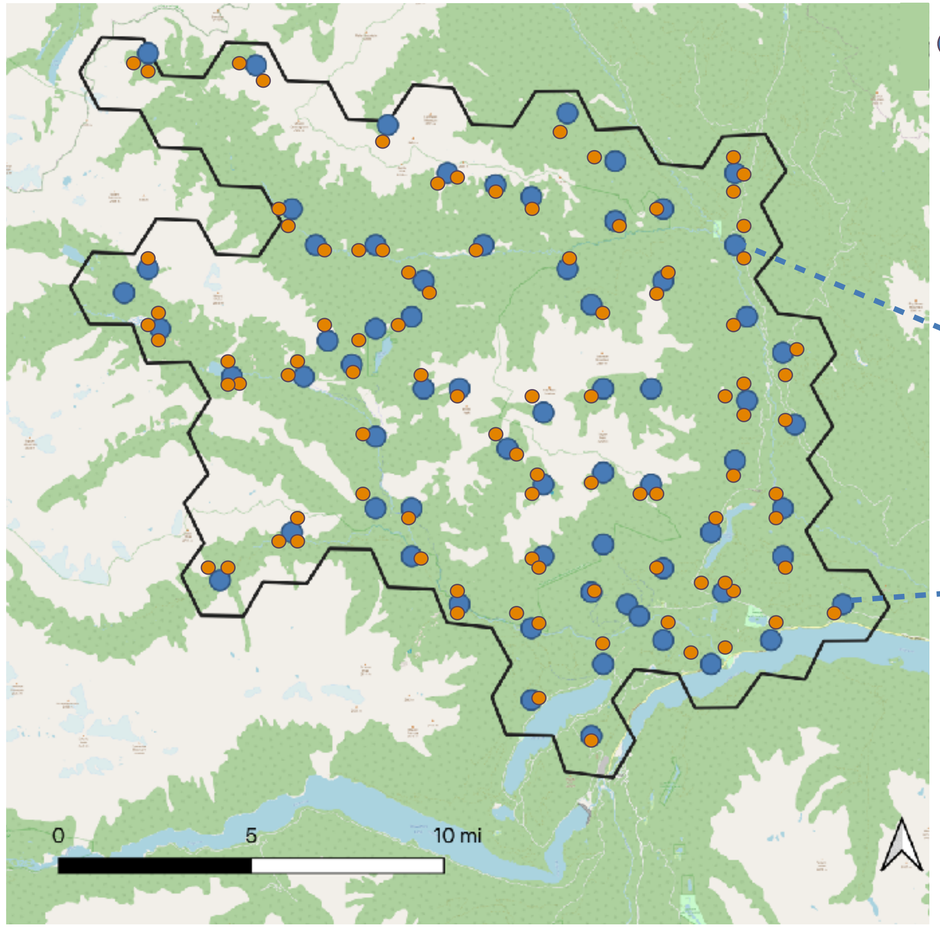


Sela Bucovetsky, 2025. Photo from iNaturalist.



Diana Lafferty, 2019. Northern Michigan University.

Camera Trapping in Biodiversity Surveys

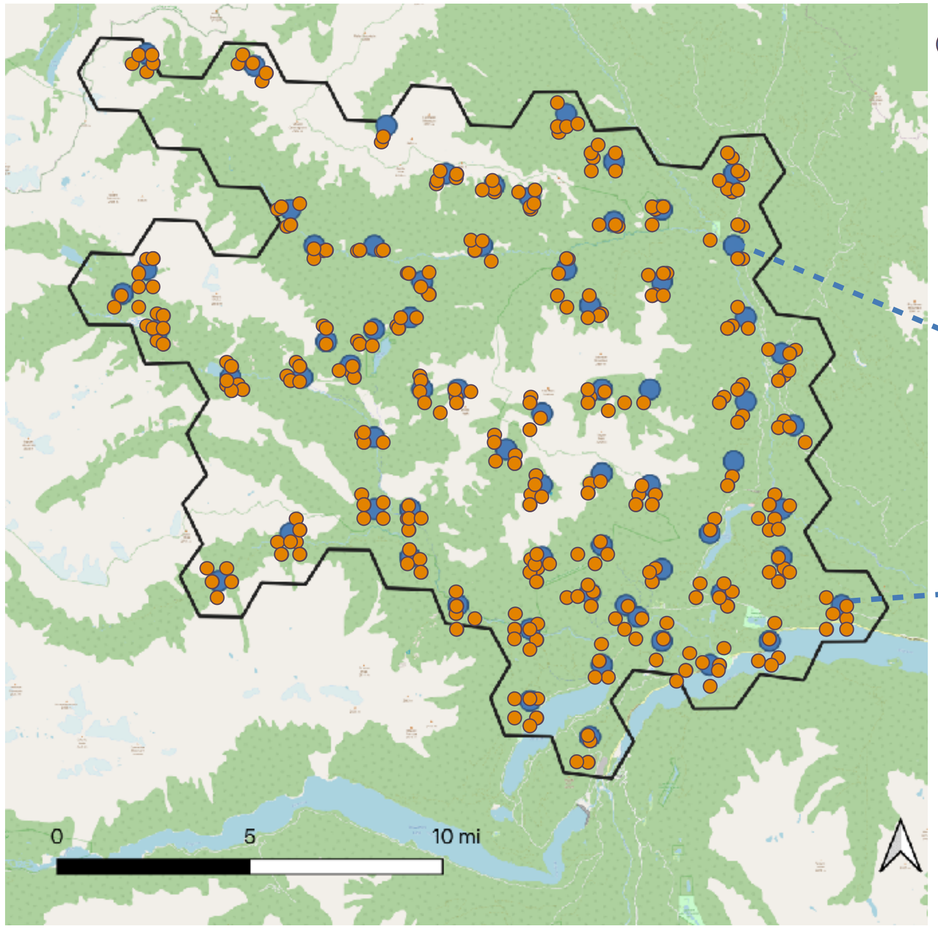


Trap
Species Observation

Animal ID				
Trap ID	A	B	C	...
1	0	0	0	
2	1	1	0	
3	0	1	0	
...				

Capture Histories from Time Period 1

Camera Trapping in Biodiversity Surveys



Trap
Species Observation

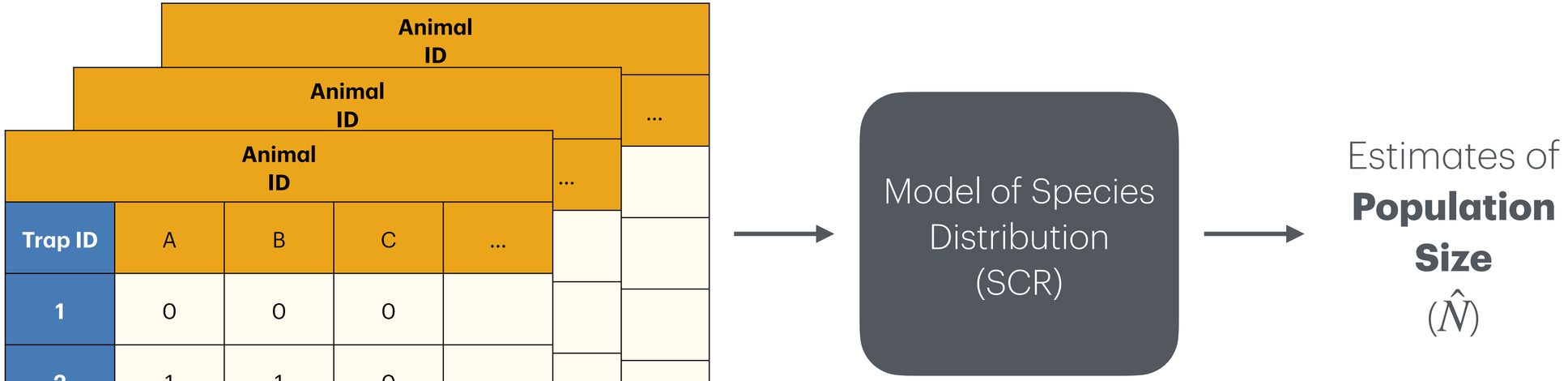
Animal ID					
Trap ID	A	B	C	...	
1	0	0	0		
2	1	1	0		
3	0	1	0		
...					

... K

2

Capture Histories from Time Period 1

Camera Trapping in Biodiversity Surveys

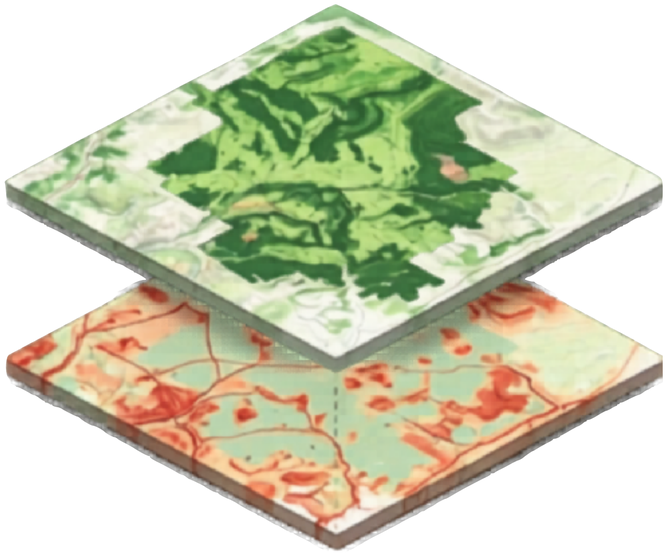


Capture Histories from Time Period 1

Spatial Capture-Recapture Models

Animal detection probability depends on their home base and habitat preferences.

Probability of Activity Center

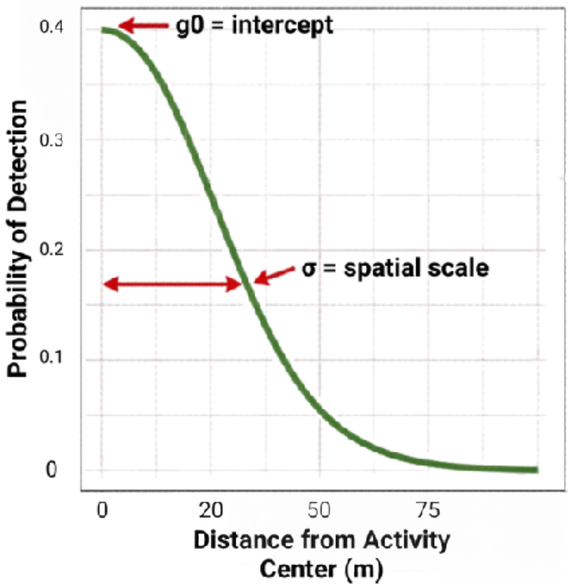


Habitats impact likelihood animal is there to begin with.

$\beta_1 =$ Percent Tree Cover
$\beta_2 =$ Human Footprint Index

+

Probability of Detection

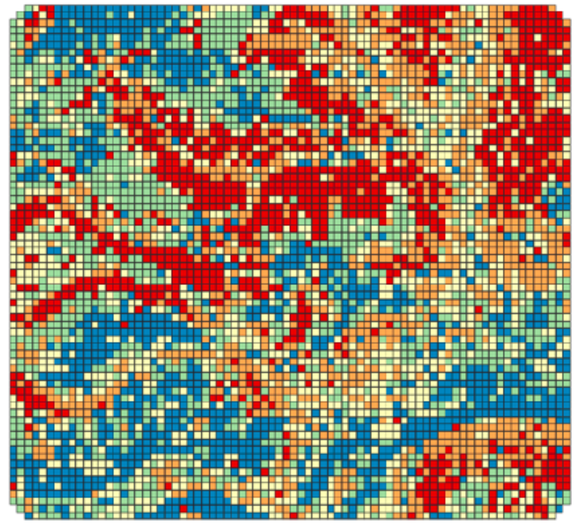


SCR models assume the likelihood of capture depends on how far that camera is from their home.

$\sigma =$ Spatial Scale
$g_0 =$ Baseline Detection Probability

→

Species Density Estimates

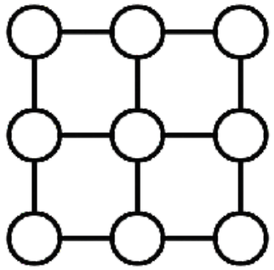


SCR models predict per-cell species density estimates D.

$\hat{N} = \sum D =$ Species Population Size

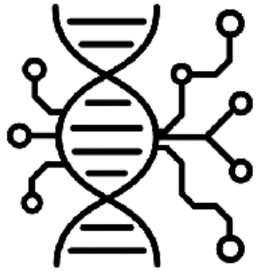
Existing Methods for Optimizing Sensor Placement

Uniform Deployment



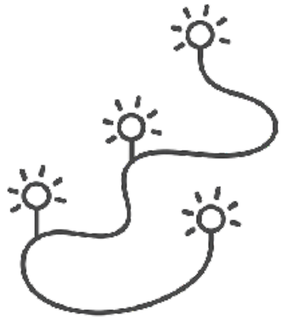
Zero knowledge

Genetic Algorithm



Assume one **known point estimate of the parameters**

Our Methodology



Considers a **range of plausible parameter values:**
Sample Average Approximation Optimization Approach

Perfect knowledge of $\beta_1, \beta_2, \sigma,$ and g_0 is **unlikely**. If the ecologists' assumed parameters turn out to be incorrect, the optimized layouts will fail to produce reliable population estimates.

Methodology: Sample Average Approximation

Sample from Uncertain Parameter Set & Simulate Capture Histories



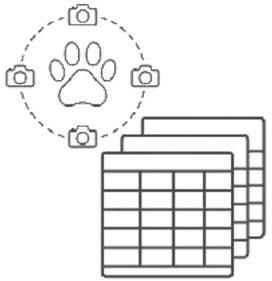
Generate 300 Representative Samples & Split into Sets A & B. If the parameters were true, what observations would we see?

Optimize for Parameter Set A



Optimize using SAA:
Greedy
or
Genetic

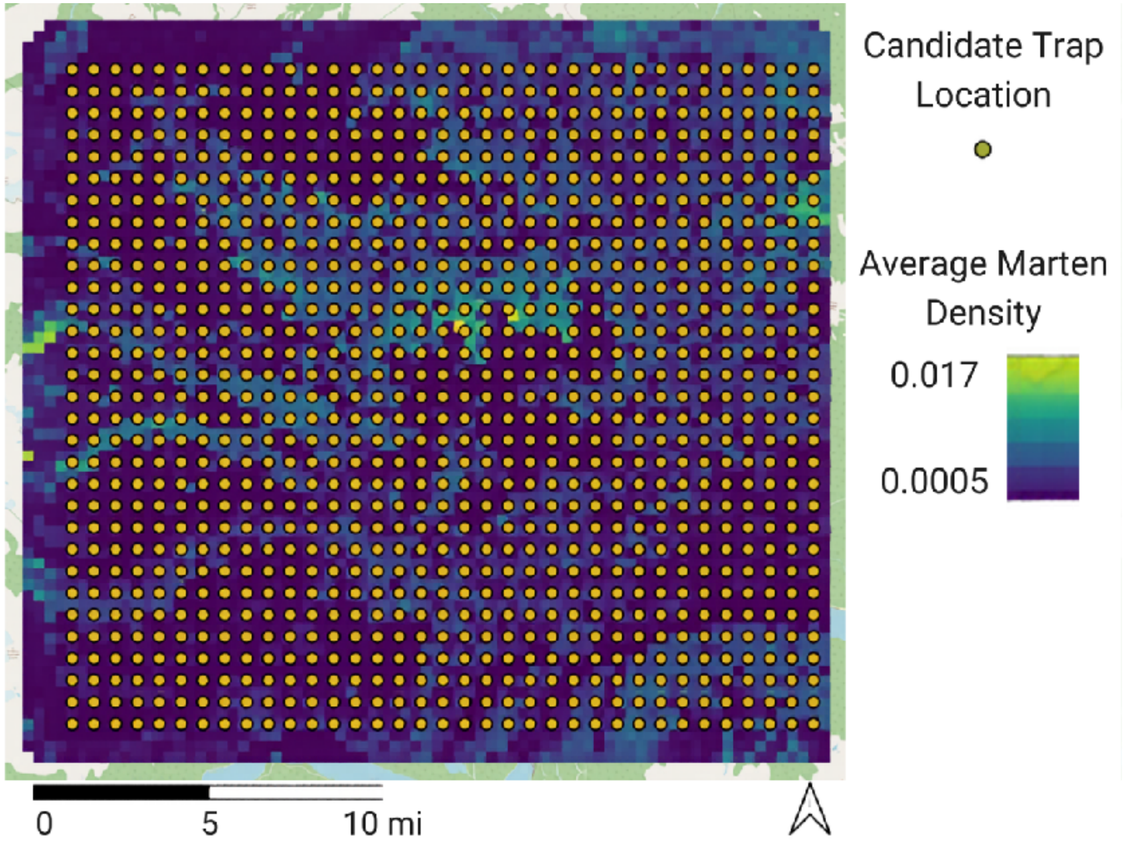
Evaluate on Parameter Set B



Estimate Population Size using SCR Models

Case Study: American Martens in Canada





Given $L = 1{,}023$ Candidate Locations x

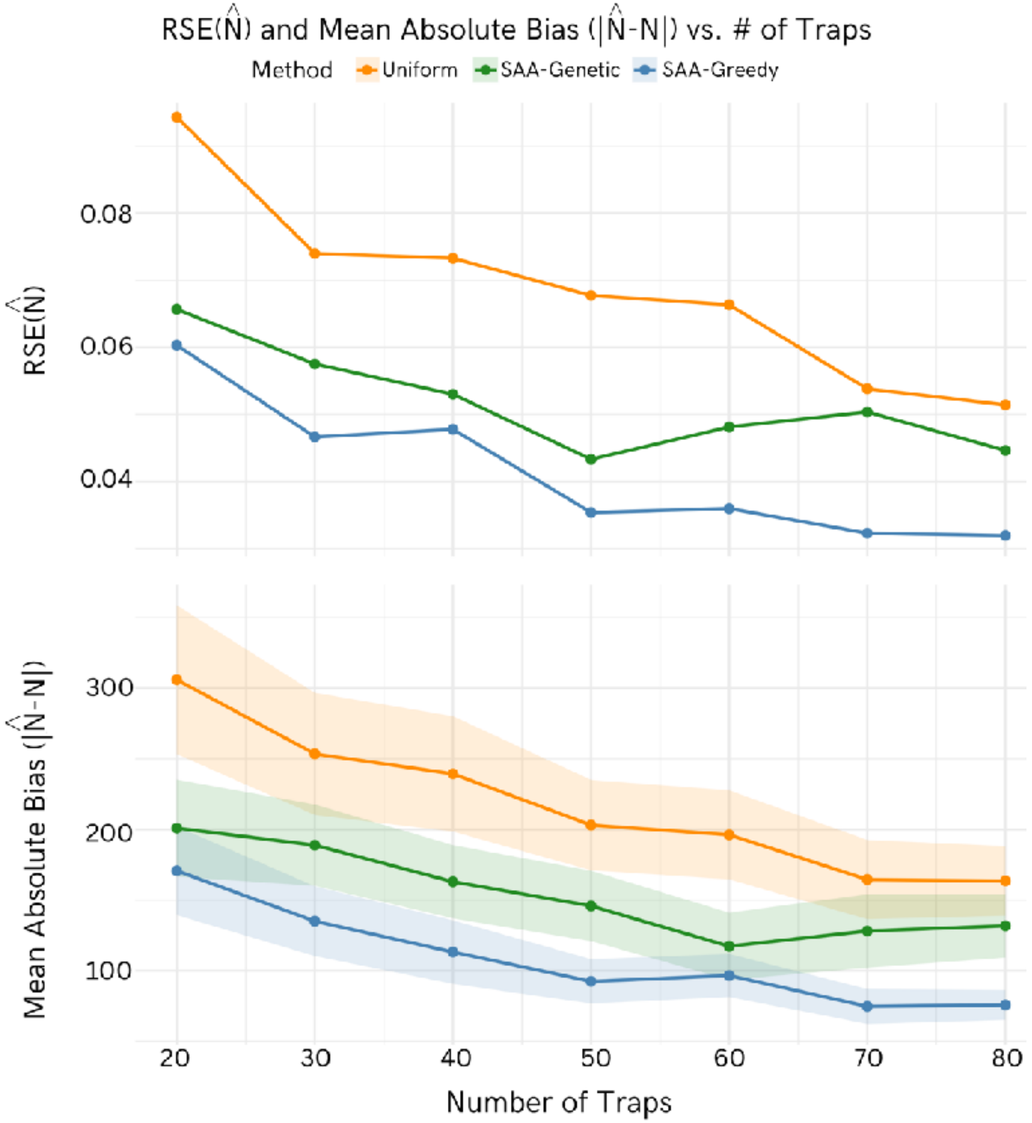
Budget $B \in [20,...,80]$

Select Best Layout Among $\binom{L}{B} \forall B \in [20,...,80]$

Such that $\sum x = B$

Results





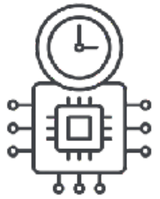
Our **SAA-Greedy** and **SAA-Genetic** methods had statistically significantly lower RSE and mean absolute bias than the **uniform** baseline across all budgets.



Mean Absolute Bias in Marten Population Estimates
↓ 30-50% better



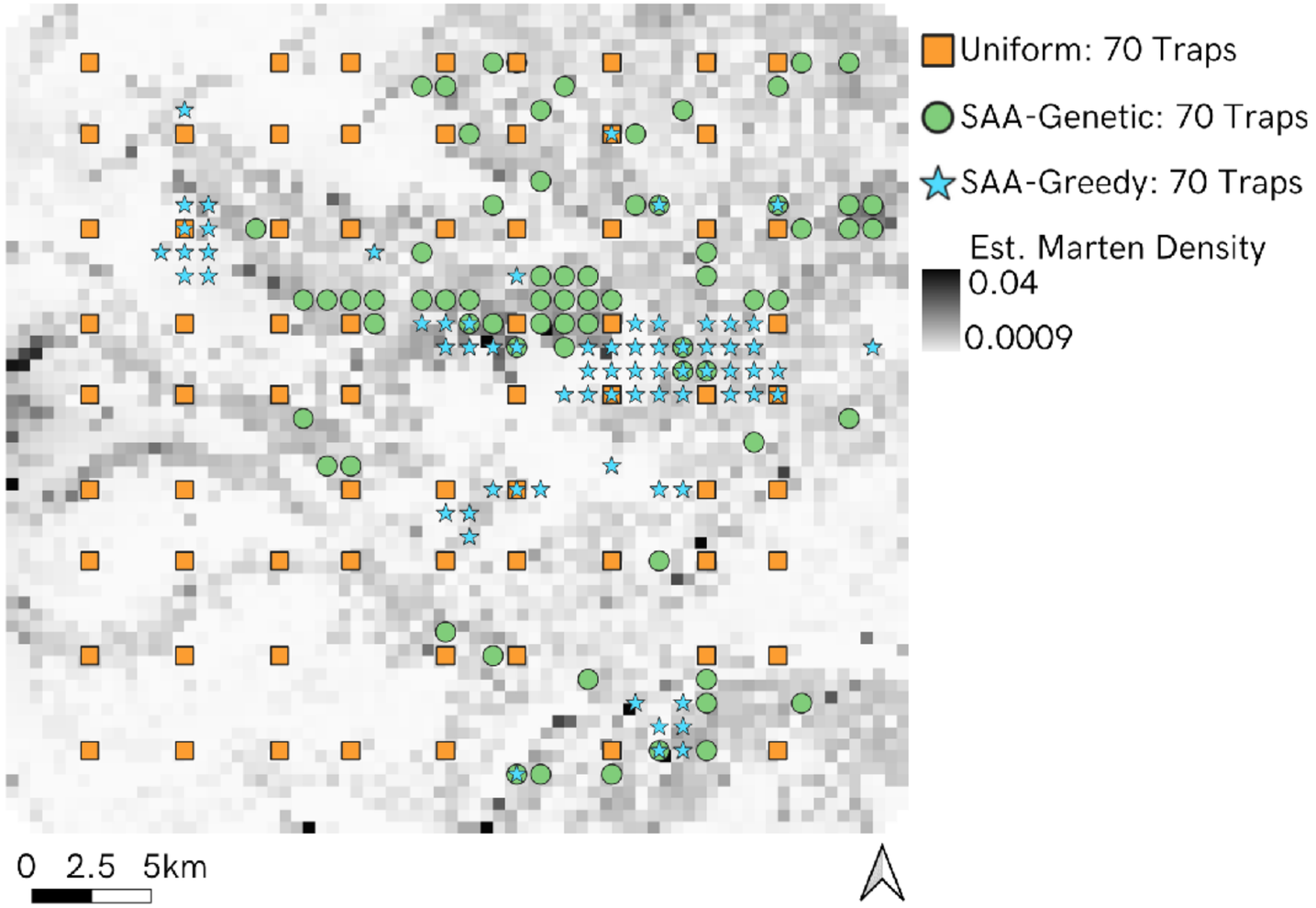
RSE in Marten Population Estimates
↓ 25-35% better



SAA-Greedy Runtimes Compared to SAA-Genetic
↓ 2.25x faster

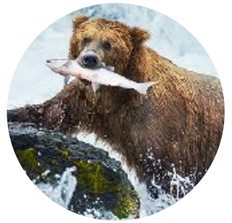
70 Camera Configurations





70 camera **uniform** configuration (orange squares), optimized modified **genetic** (green circles), and optimized **greedy** (blue stars). Clustering in algorithmic methods follows the habitat preferences of target species.

Case Study: Grizzly Bears in Canada

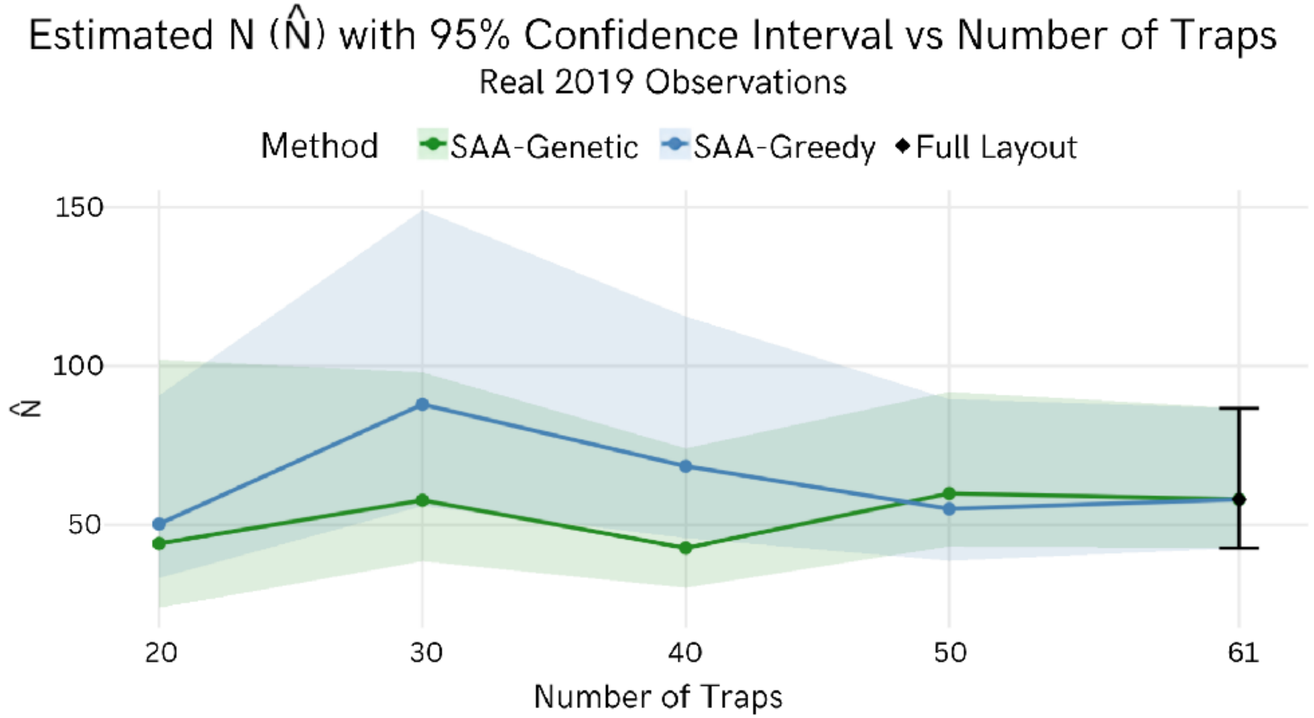


Can we shrink an existing camera network without losing precision?

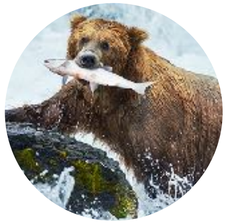
A 10-year grizzly bear monitoring program in British Columbia uses 61 cameras. Which cameras could we remove without impacting performance?

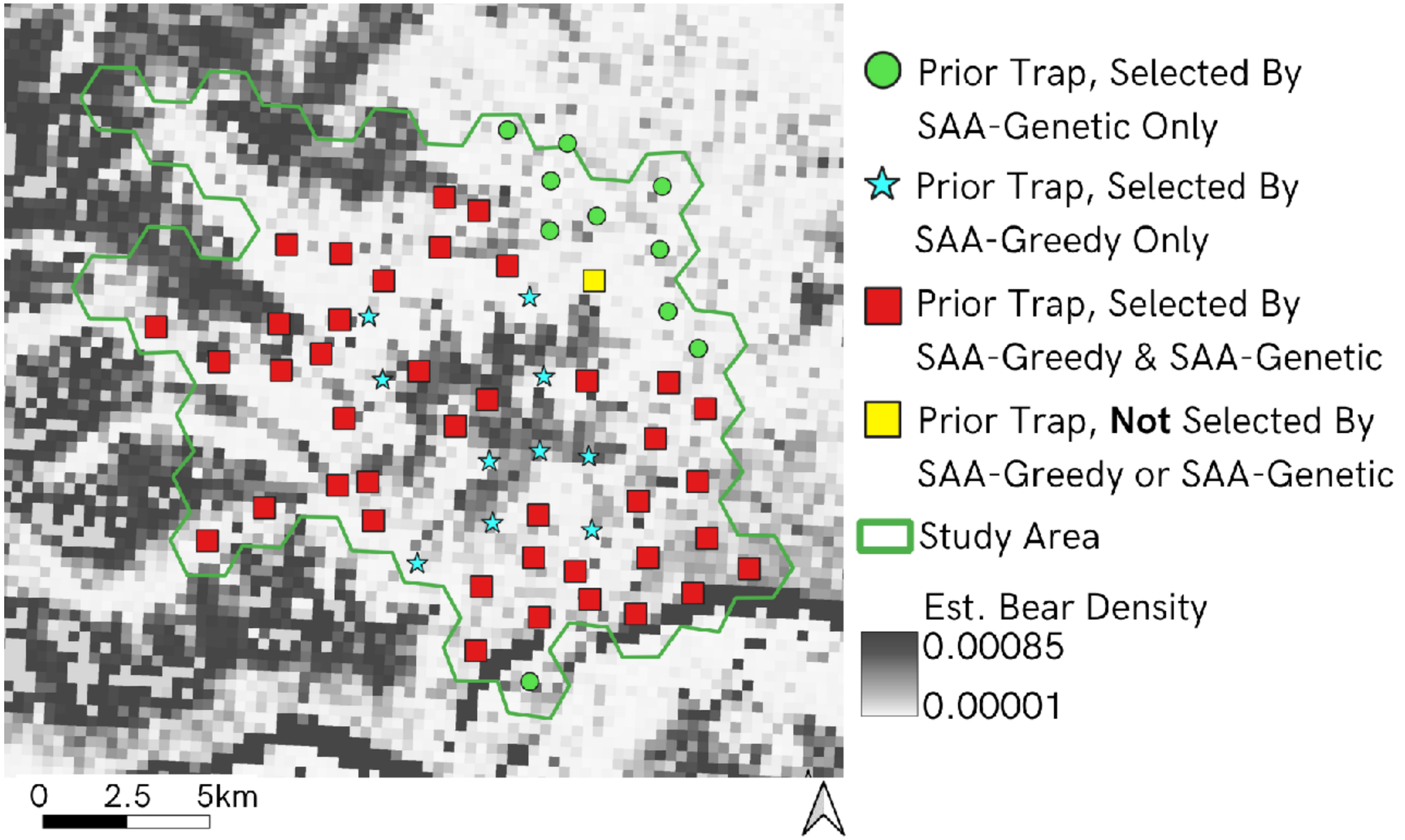
Results

- 18% fewer cameras can be removed with little to no impact on the estimated population size.
- Any fewer cameras, SAA-Greedy tends to overestimate and SAA-Genetic tends to underestimate.



50 Camera Configurations





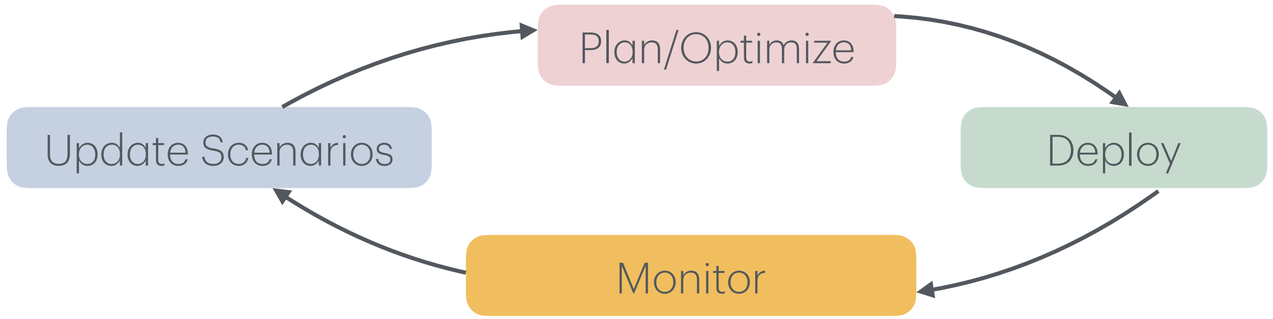
Where do we go from here?

What did we learn?

Using prior knowledge on how species respond to the environment to optimize sensor placement is better than a uniform approach.

What's next: adaptive sensor management

Monitoring programs can run for years. As climate change shifts species ranges and habitats, we can use our method across seasons, deciding which sensors to move as the scenarios shift with new field data and climate projections.



There is a need for sequential planning techniques to support this effort.

Thank you!



hannahmu@usc.edu

CONSERVATION SCIENCE PARTNERS

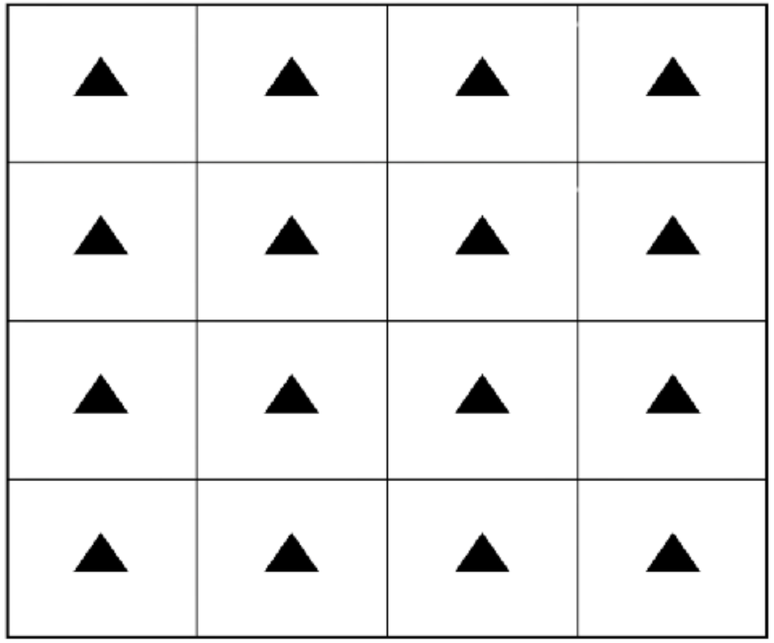
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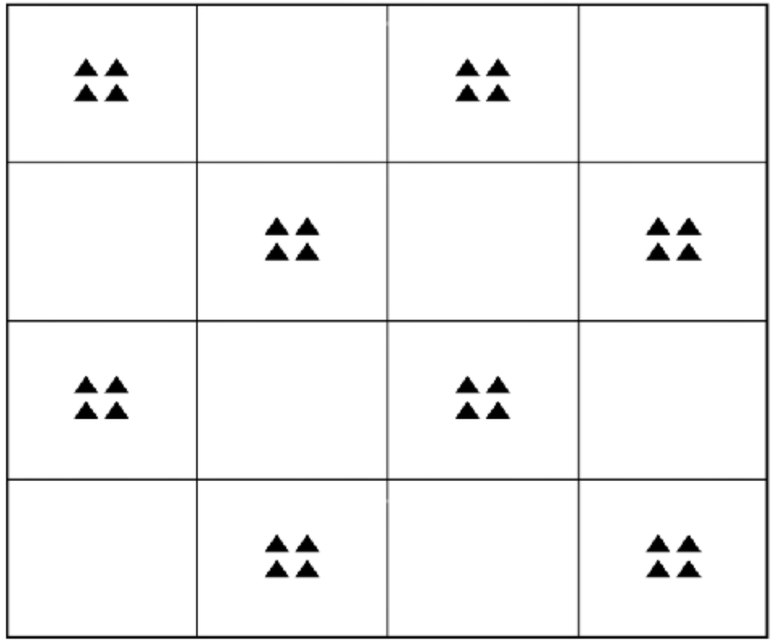
Appendix

Camera Trapping in Biodiversity Surveys

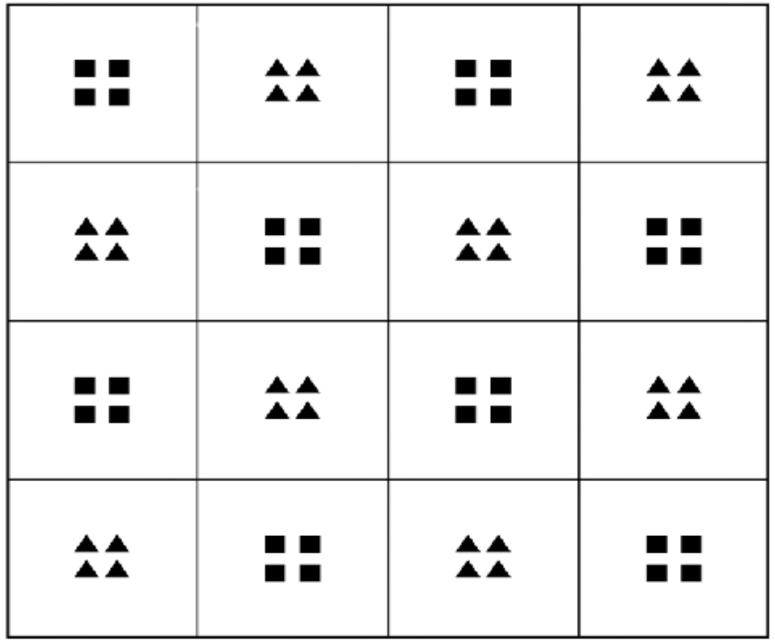
Uniform: Evenly space sensors across the landscape.



Clustered: Group cameras in key locations within the grid.



Sequential: Group cameras, but move them over time.



Opportunistic: Cameras are placed along hiking trails or roads because they are easier for humans to reach.

Can we do better?

Spatial Capture-Recapture Models

For each parameter scenario $\alpha = (g_0, \sigma, \beta_1, N, \dots)$ and capture history H_α...

Compute Likelihood - $L(\alpha, H_\alpha)$

How likely are we to truly observe all the capture histories if the parameter values represent the ground truth?

Maximum Likelihood Estimation - $\max L(\alpha, H_\alpha)$

Maximize the likelihood with respect to parameter values to obtain $\hat{g}_0, \hat{\sigma}, \hat{N} \dots$

Measure Performance

Bias: $\hat{N} - N$

Relative Standard Error: $RSE(\hat{N})$

...

Average performance across all parameter scenarios (Mean Bias, Mean RSE)

Stochastic Optimization Problem Formulation

Given

α = vector of detection and habitat parameters $g_0, \sigma, D, \ldots$

$\mathcal{J} = \{1, \ldots, J\}$ = finite set of candidate camera trap locations

B = sensor budget

Find

$\mathbf{x}^\star \in \{0,1\}^J$ with $\sum_{j=1}^{J} x_j = B$ that minimizes loss f, where $f = RSE(\hat{N})$ as output from SCR.

Objective

$Z(\mathbf{x}) = \mathbb{E} f(\mathbf{x}; \alpha)$ where α is a random element and $f(x; \alpha)$ is a (deterministic) real-valued function.

Making the stochastic program $\mathbf{x}^\star \in \arg \min_{\mathbf{x} \in \{0,1\}^J} Z(\mathbf{x})$ subject to $\sum_{j=1}^{J} x_j = B$.

Sample Average Approximation

A common method for transforming stochastic optimization problems into a deterministic analogue.

Consider a set of G iid parameter combination samples $\{\boldsymbol{\alpha}^{(g)}\}_{g=1}^{G}$

Loss Function

Loss becomes the average relative standard error of the estimated population size $\frac{1}{G}\sum_{g=1}^{G} RSE\left(\hat{N} \mid \mathbf{x}, \boldsymbol{\alpha}^{(g)}\right)$ across each plausible $\boldsymbol{\alpha}$ given selected sensor locations $\mathbf{x}$.

Sample Average Approximation

A common method for transforming stochastic optimization problems into a deterministic analogue.

Consider a set of G iid parameter combination samples $\{\alpha^{(g)}\}_{g=1}^{G}$

Sampling for Optimization

Repeatedly solve M finite-sample approximations of the optimization. In run $m = 1,\ldots, M$, draw a training set of N parameter scenarios $\alpha^{(m,1)}, \ldots, \alpha^{(m,N)}$ from the simulation pool G (with replacement) and solve

$$\mathbf{y}^{(m)} \in \arg\min_{\mathbf{x}\in\{0,1\}^J} \hat{Z}^{(m)}(\mathbf{x}) \quad \text{subject to} \quad \sum_{j=1}^{J} x_j = B$$

Where the sample average objective is

$$\hat{Z}^{(m)}(\mathbf{x}) = \frac{1}{N}\sum_{n=1}^{N} RSE\left(\hat{N} \mid \mathbf{x}, \alpha^{(m,n)}\right)$$

Sample Average Approximation

A common method for transforming stochastic optimization problems into a deterministic analogue.

Consider a set of G iid parameter combination samples $\{\alpha^{(g)}\}_{g=1}^{G}$

Validation using SCR

Evaluate each $\mathbf{y}^{(m)}$ directly using SCR on a separate validation set of unseen parameter scenarios in G by recording the average $RSE(\hat{N})$ across those draws.

Testing using SCR

Select the $\mathbf{y}^{(m)}$ that achieves the lowest $RSE(\hat{N})$ on the validation runs, and test it on another unseen set of parameters in G using SCR, constituting the test results.

Fast Computation of Population Size Precision

$$\overline{RSE(\hat{N})} = \frac{1}{\sqrt{(\min(E_n, E_r))}}$$

E_n = Expected Number of Unique Individuals Detected

$$= \sum_l D(l) \cdot \Pr[\text{detect individual with activity center } l]$$

$$= \sum_l D(l) \cdot \{1 - \Pr[\text{never detect individual with activity center } l]\}$$

$$= \sum_l D(l) \cdot \left\{ 1 - \prod_{j=1}^{J} (1 - x_j \cdot \Pr[l, j; \alpha])^K \right\}$$

E_r = Expected Number of Recaptures

= Expected Number of Captures $- E_n$

$$= \sum_l \sum_j \sum_k D(l) \cdot x_j \cdot \Pr[l, j, \alpha] - E_n$$

Probability individual with activity center l, spotted at trap j, under parameter sceanrio α

$$= g_0 \cdot e^{-(\frac{1}{2\sigma^2}) \cdot d(l,j)}$$

Binary decision if trap activated at location j

Prior of density of target species in l

Murray Efford & John Boulanger. Fast evaluation of study designs for spatially explicit capture–recapture. *Methods in Ecology and Evolution*. 2019

Forward Greedy

Algorithm 1: Forward Greedy Algorithm for Trap Selection

$\mathbf{x} = [0, 0, \ldots, 0]$ (no traps selected)
for $s = 1$ to budget **do**
  $min_{RSE} \leftarrow \infty$
  $add \leftarrow$ None
  for each trap j with $x_j = 0$ **do**
    Temporarily set $x_j = 1$ to get $\mathbf{x}_{temp}$
    Evaluate objective $RSE(\mathbf{x}_{temp})$ across all scenarios
    if $RSE(\mathbf{x}_{temp}) < min_{RSE}$ **then**
      $add \leftarrow j$
      $min_{RSE} \leftarrow RSE(\mathbf{x}_{temp})$
    end if
  end for
  Set $x_{add} \leftarrow 1$ (permanently add trap with **largest decrease in RSE**)
end for
return $\mathbf{x}$

$$\hat{RSE}(D) = \frac{1}{\sqrt{(\min(E_n, E_r))}}$$

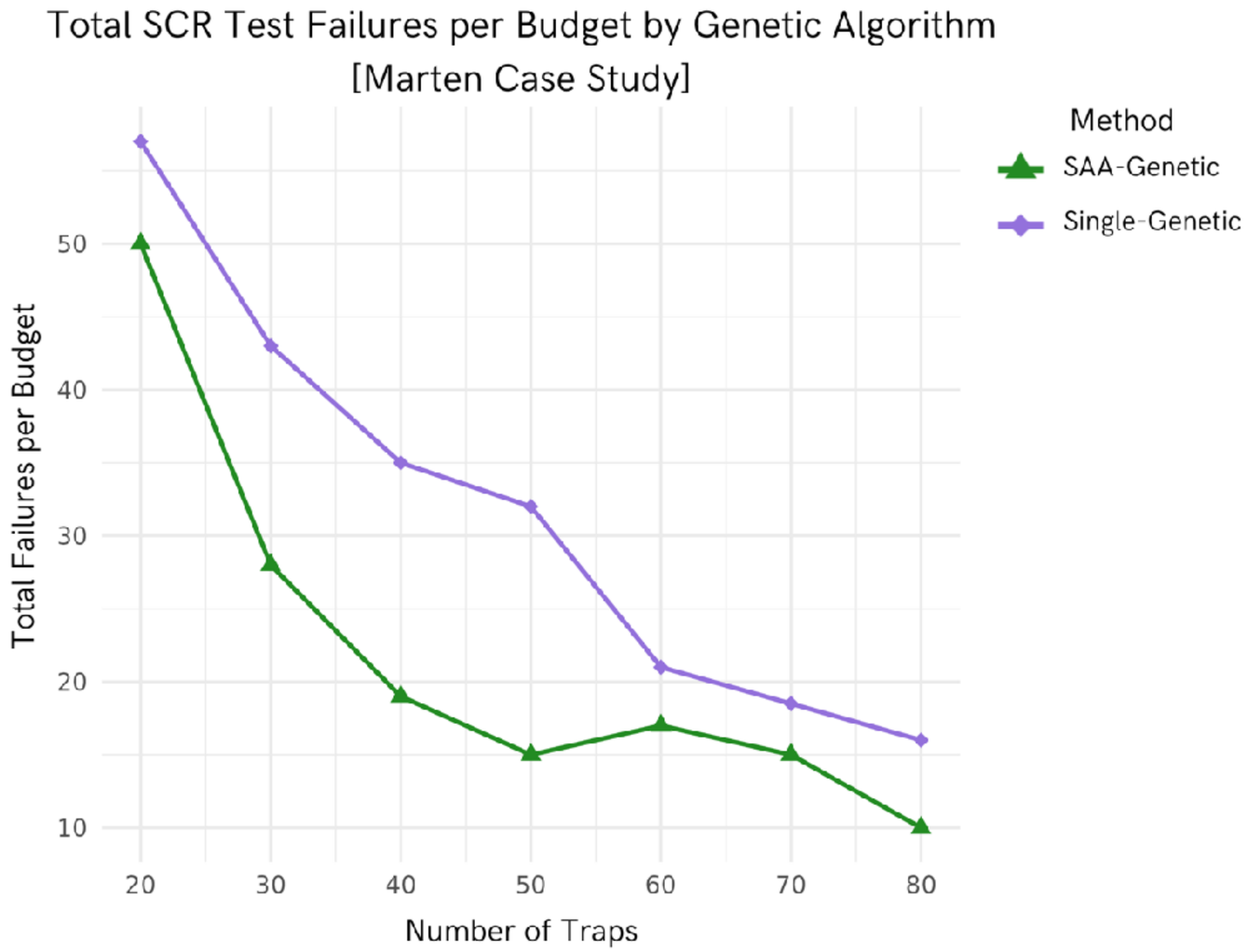
Marten Results

Number of Traps	Method	RSE($\hat{N}$)†	Mean Abs. Bias†
20	Uniform	0.086	311.84
	Genetic	0.087	207.65
	Greedy	^{UG}**0.062**	^{U}**171.51**
30	Uniform	0.070	234.58
	Genetic	0.064	199.07
	Greedy	^{U}**0.055**	^{U}**158.59**
40	Uniform	0.070	227.69
	Genetic	0.057	163.55
	Greedy	^{UG}**0.048**	^{UG}**115.33**
50	Uniform	0.068	214.21
	Genetic	0.051	159.22
	Greedy	^{UG}**0.036**	^{UG}**97.82**
60	Uniform	0.063	195.68
	Genetic	0.047	120.32
	Greedy	^{U}**0.041**	^{U}**109.20**
70	Uniform	0.052	151.98
	Genetic	0.049	128.62
	Greedy	^{UG}**0.032**	^{UG}**73.35**
80	Uniform	0.057	168.21
	Genetic	0.044	122.76
	Greedy	^{UG}**0.034**	^{UG}**81.88**

Table 2: Relative standard error RSE($\hat{N}$) and mean absolute bias $|\hat{N} - N|$ by trap budget and method. †Superscripts on greedy method rows imply statistically significantly lower RSE or bias at $\alpha = 0.05$ relative to other methods: ^{U} vs Uniform and ^{G} vs Genetic.

Results




Total SCR Test Failures per Budget by Genetic Algorithm
[Marten Case Study]
Total Failures per Budget
50
40
30
20
10
20
30
40
50
60
70
80
Number of Traps
Method
SAA-Genetic
Single-Genetic